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A Guide to Trend Following Strategies

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Trend following is one of the most popular and enduring systematic trading strategies. An attractive long-term track record and strong returns during market crises have resulted in trend following becoming a vital component in many portfolios.

In this paper we analyze the general profile of trend following strategies, and demonstrate how to build such a strategy in four steps.

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What Is A Trend Following Strategy

A trend following strategy aims to systematically capitalize on persistent upward and downward trends in asset prices. Investors who employ a trend following strategy analyze price data and buy an asset when its price is rising and sell (or short) when the price is falling. This approach can be applied to a wide range of asset classes and instruments to benefit from diversification across a broad investment universe. Investors allocate long and short positions to futures and forwards for rapid and cost-efficient implementation.

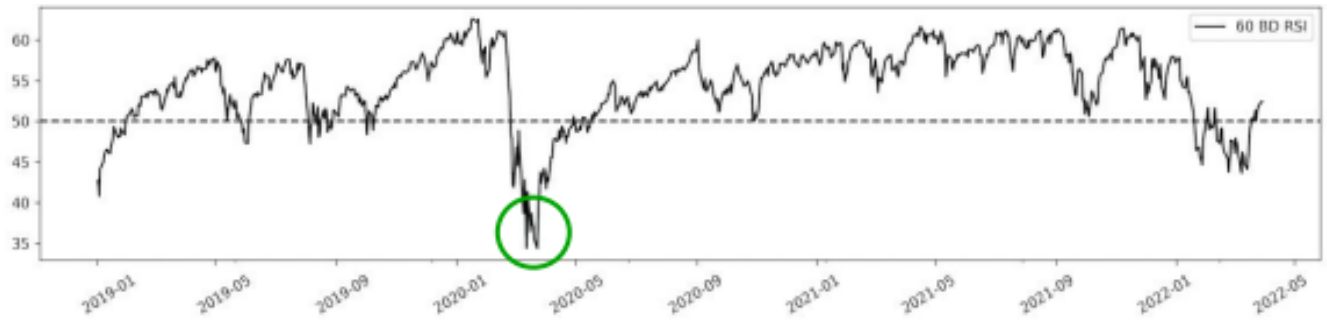
Trend Indicators And Position Sizing

Trends may be detected using myriad techniques ranging from traditional technical analysis and statistical time-series filtering to utilizing machine learning and AI. The choice of which to apply is up to the individual investor's investment philosophy and preferences. Below we present a subset of the most commonly used trend indicators.

Relative Strength Index (RSI)

RSI is a momentum oscillator that measures the speed and change of price movements, and is used to spot overbought and oversold market conditions. The data is normalized to move within a range of 0 to 100. In this example, we calculate RSI for the E-Mini S&P 500 futures using a lookback window of 60 trading days. Exhibit 1 for instance indicates an oversold situation in April 2020.

Exhibit 1: Relative strength index of the S&P 500 futures

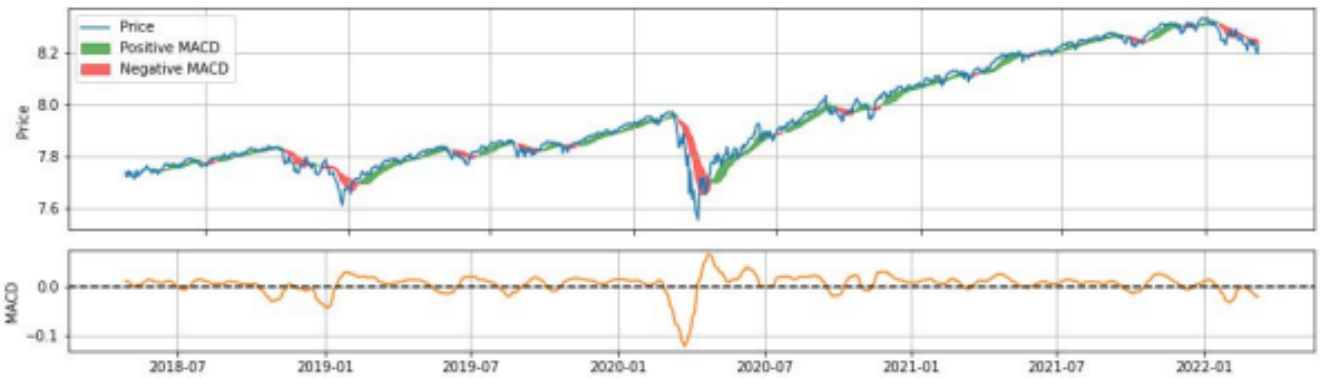


Source: SigTech analysis; calculations based on data for the time period January 2019 to May 2022

Moving Average Convergence Divergence (MACD)

MACD is used to measure the direction and the strength of a trend. This indicator compares the difference of the moving averages calculated over different windows. In our example for the E-Mini S&P 500 futures, we use the past 26 trading days to calculate the long EMA, and 12 days for the short EMA.

Exhibit 2: Moving average convergence divergence of the S&P 500 futures



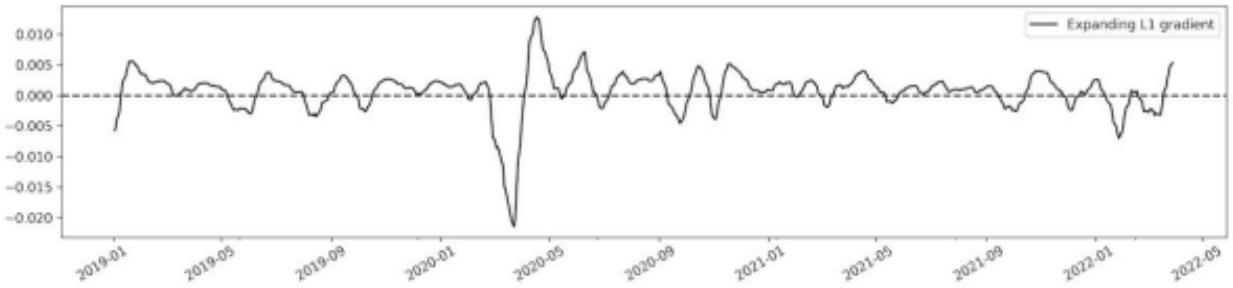
Source: SigTech analysis; calculations based on data for the time period May 2018 to May 2022

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Filters

Various statistical methods can be used to filter out high frequency noise from the underlying value of a time series. A simple average can be viewed as the most basic filter as it takes the average values in a period to estimate the underlying value. More advanced filters such as L1 and Hodrick-Prescott try to identify the time series' trend by minimizing noise and applying a penalty associated with variations in the estimated trend. A measure of direction can be obtained from the filtered series by evaluating the gradient. Exhibit 3 shows an L1 filter for the E-Mini S&P 500 futures.

Exhibit 3: L1 filter of the S&P 500 futures



Source: SigTech analysis; calculations based on data for the time period January 2019 to May 2022

To convert trend signals to portfolio positions, the two most common approaches are inverse volatility and risk parity. Inverse volatility results in a portfolio where the asset weights are set inversely proportional to its volatility. Whereas risk parity uses various risk measures (e.g. volatility, covariances) to ensure an equal risk contribution from each asset. In addition, strict risk management processes that define stop-losses and exposure limits have a large impact on position sizing.

The Rationale of Trends

One key rationale behind why trends exist and how investors can exploit these to systematically achieve strong investment results is found in behavioral finance. By studying the impact of psychology on the behavior of market participants and the ensuing effect on asset prices, behavioral finance seeks to resolve contradictions between classical economic theory and reality. It has identified various cognitive biases that explains how and why investors act on information in an - at least in hindsight - irrational manner. Examples of such biases that help explain the existence of trends are:

- Herding: investors tend to follow actions of other investors
- Overreaction: investors' reaction to new information is exaggerated
- Confirmation bias: investors actively seek and interpret information to back up existing beliefs

A further explanation for the appearance of trends is the non-instantaneous dissemination and reception of information, where news and data require time to be incorporated into market prices.

Historical Performance Overview

Trend following has been an investment strategy for as long as investments have been made, and has enjoyed a strong standing in the investment community since the 70s. Investors' strong demand for trend following strategies is driven by its attractive long-term track record and its positive diversification properties.

Historically, trend-based strategies have not only exhibited a low correlation to traditional asset classes, but also to other hedge fund strategies. Furthermore, its long volatility profile (i.e. it generally profits in periods of rising volatility) has resulted in strong returns during market crashes such as in 2008.

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Exhibit 3: Risk/return overview

Risk/return overview 2000-2022*

	Trend following	Equities	Bonds	Commodities	Hedge funds
Return, p.a.	5.3%	5.4%	3.8%	0.5%	2.9%
Volatility	13.4%	15.6%	6.4%	24.1%	5.5%
Downside volatility	8.5%	8.3%	4.2%	12.3%	3.0%
Upside volatility	7.9%	12.4%	3.9%	19.6%	5.2%
Return / risk	0.40	0.35	0.59	0.02	0.53
Max. drawdown	-20.7%	-54.0%	-10.3%	-87.2%	-25.2%

Correlation matrix*

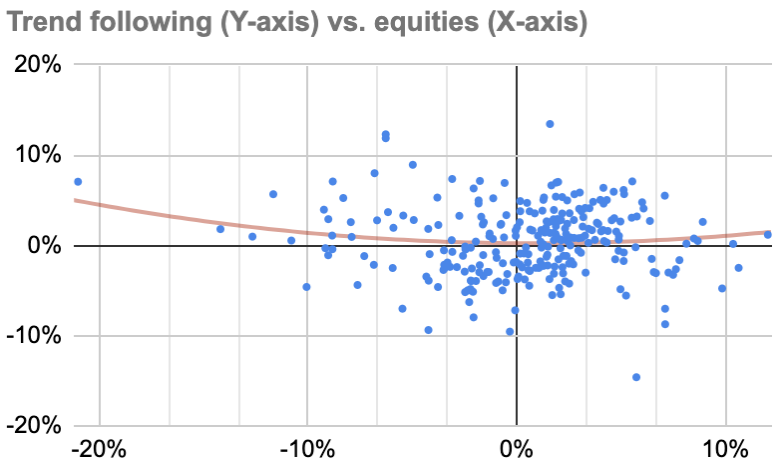
	Trend following	Equities	Bonds	Commodities	Hedge funds
Trend following	1				
Equities	-0.05	1			
Bonds	0.25	0.16	1		
Commodities	0.10	0.44	0.12	1	
Hedge funds	0.25	0.70	0.14	0.54	1

Source: SigTech analysis; calculations based on data for the time period January 2000 to May 2022. Societe Generale, Bloomberg Calculations based on USD returns for the time period January 2000 until January 2022. Trend following = SG Trend Index, Equities = MSCI World Net Total Return index, Bonds = FTSE World Government Bond Total Return index, Commodities = S&P GSCI Total Return index, Hedge funds = HFRX Global Composite index

Also noteworthy is the strategy's positive skewness - i.e. positive returns are on average higher in magnitude than negative returns. The reason being that trend following strategies have a natural tendency to close out losing positions, while letting "winners run".

Exhibits 4-7 compare the monthly returns of trend following strategies with equities, bonds, commodities and hedge fund index returns. The results show a pronounced "smile", i.e. indicating that trend following performs particularly well when other asset classes experience larger swings. The strong relative performance of trend following in down markets has led investors to coin the strategy "crisis alpha". On the flip side, trend-based strategies do - not surprisingly - experience difficulties in range bound markets, when no clear trend persists.

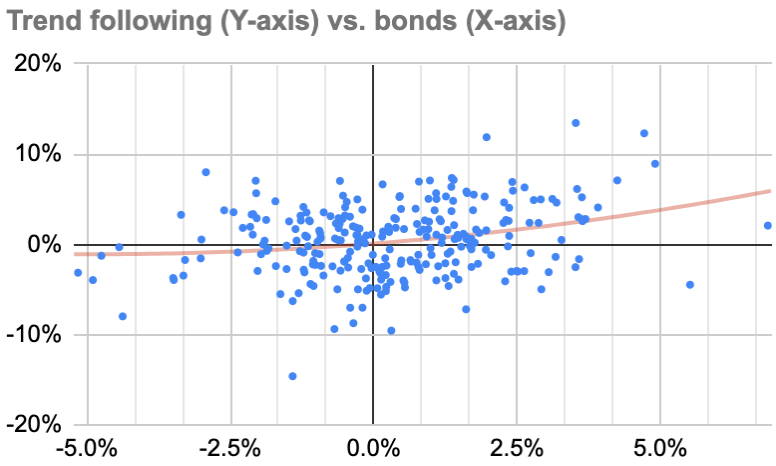
Exhibit 4: Volatility of trend following vs. equities



Note: SigTech analysis; calculations based on USD returns for the time period January 2000 until January 2022. Trend following = SG Trend Index, Equities = MSCI World Net Total Return index

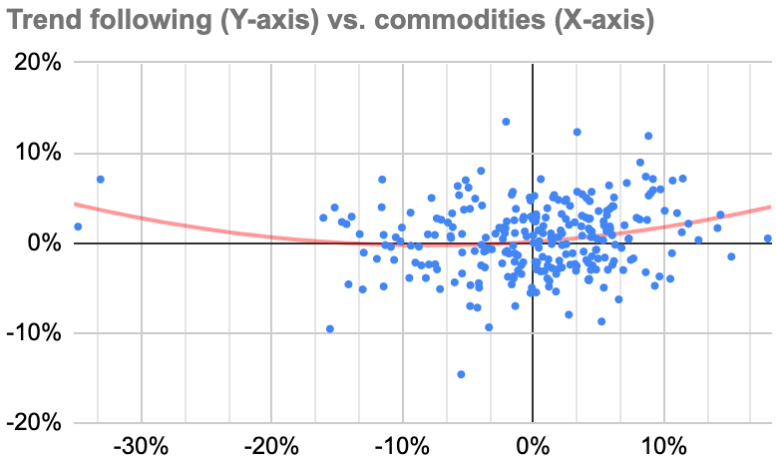
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Exhibit 5: Volatility of trend following vs. bonds



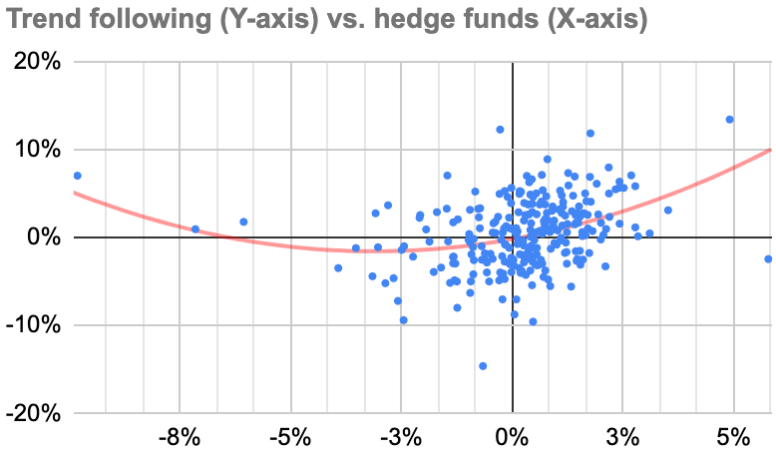
Note: SigTech analysis; calculations based on USD returns for the time period January 2000 until January 2022. Trend following = SG Trend Index, Bonds = FTSE World Government Bond Total Return index

Exhibit 6: Volatility of trend following vs. commodities



Note: SigTech analysis; calculations based on USD returns for the time period January 2000 until January 2022. Trend following = SG Trend Index, Commodities = S&P GSCI Total Return index

Exhibit 7: Volatility of trend following vs. hedge funds



Note: SigTech analysis; calculations based on USD returns for the time period January 2000 until January 2022. Trend following = SG Trend Index, Hedge funds = HFRX Global Composite index

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How to Build A Trend Following Strategy

In this chapter, we will go through a four step process on how to build a bespoke trend following strategy.

1. Investment universe

As a first step, we define our bespoke investment universe. For this strategy, we select 52 highly liquid futures across the asset classes equities, fixed income, FX and commodities. To prepare the backtesting, we create historical rolling futures time series for each asset using cleaned and validated data. With the objective to minimize the disconnect between backtesting and live trading results, we make use of an event-driven backtesting engine that allows for detailed customization of, for example, what trading commission to apply, and what strategy is implemented to roll from one future to the next. The time series are created for the entire backtesting period that starts in 2002.

2. Trend detection

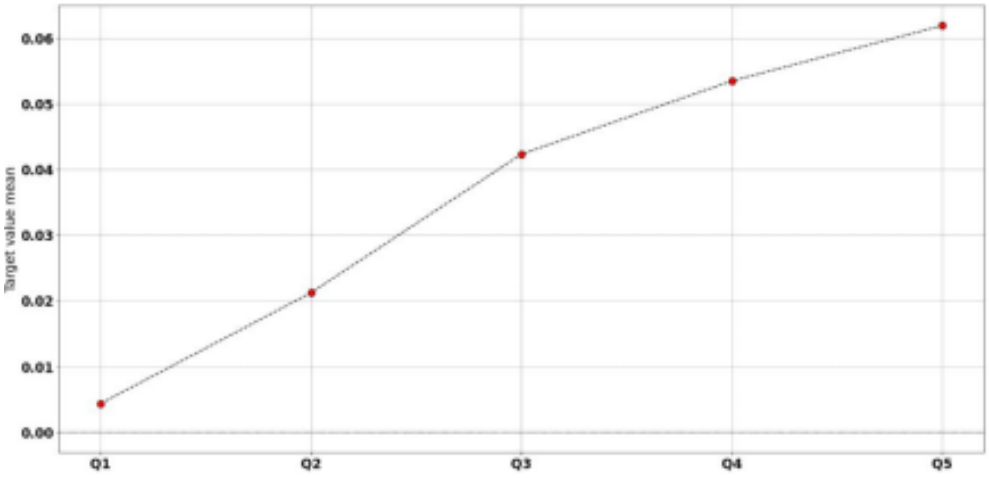
In our example, we apply an indicator of directional trend that calculates the average direction of the time series based on a 60 trading day window using the following formula:

$$\frac{(\text{number of daily upward returns} - \text{number of daily downward returns})}{\text{total number of days}}$$

The values are scaled based on extreme observations over a rolling three year window, resulting in an indicator showing values between -1 and 1.

To showcase the effectiveness of this trend measure, the following extract from our platform's signal analysis report presents how well the current unscaled signal predicts the next day's directional move. The signal values are put into quantile buckets on the x-axis and the upward sloping line shows that the stronger the trend signal, the more likely it is to continue over the next period. This confirms that - at least in the in-sample period - the trend signal can be used to generate alpha.

Exhibit 8: Signal quality



Source: SigTech Alpha Report

3. Portfolio construction

The next step is to use the trend signals and convert them into a tradable investment strategy taking long and short positions in the various assets. To determine the portfolio weight of each asset, we apply an inverse volatility methodology, i.e. assets are weighted in inverse proportion to their risk. To calculate the risk measure we use an exponentially weighted historical volatility estimate, with a 60 day span. After we have balanced the volatility of each asset in the portfolio, the overall leverage of the portfolio is a function of targeting a long-term volatility of 5% based on a 60 day covariance estimate.

4. Backtesting

The last step is to run the backtest using settings that mimic our live trading environment. We rebalance weekly, apply a 2% threshold to trigger a rebalancing and a bid-ask spread of 5 bps.

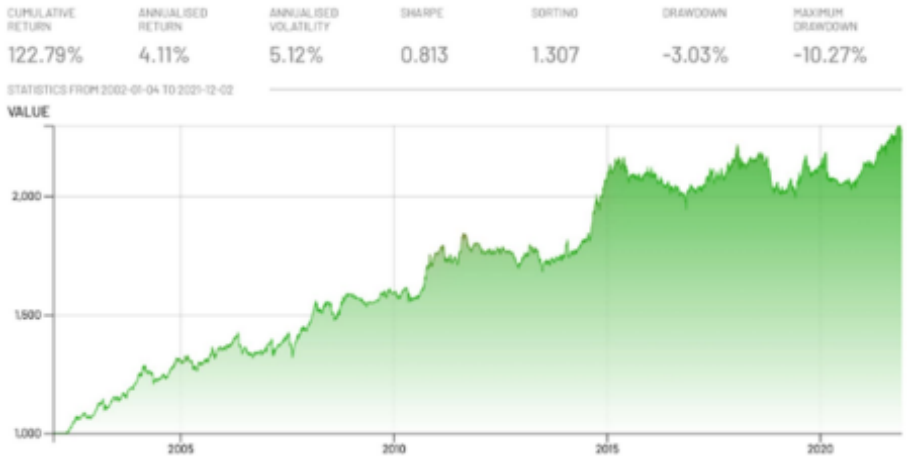
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What do the results look like?

Over the past 20 years, the trend following strategy achieved an average net return of 4.1% (5.0% gross) , although the performance over the last couple of years has, in line with trend following strategies in general, been weaker. From a risk perspective, the strategy's realized volatility is 5.1% - which is very close to the targeted volatility of 5% - and its maximum drawdown is -10.3%. The strategy's Sharpe ratio is 0.81.

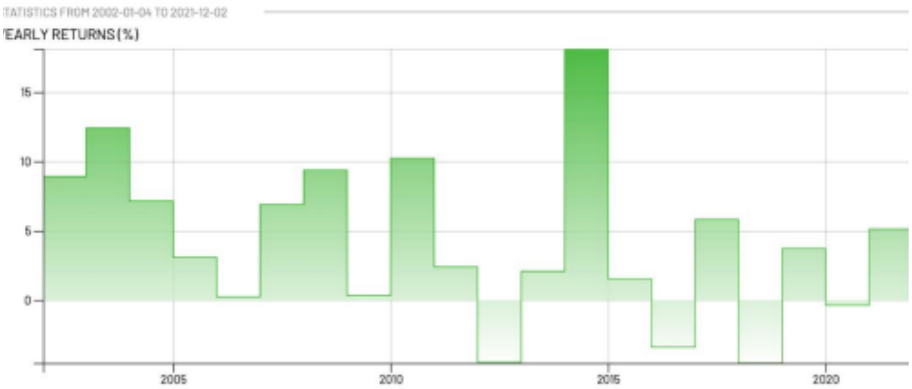
Analyzing the yearly returns, what sticks out is that the strategy only exhibits four down years (averaging -3.2%) over the entire 20-year backtesting period, and offers attractive diversification in years when equities performed poorly (e.g. 2002 and 2008). As a further reference point, this trend following strategy exhibits an overall strong average correlation of 0.7 to the broad SG Trend index.

Exhibit 9: Historical performance



Source: SigTech analysis, calculations based on USD returns for the time period January 2000 until January 2022

Exhibit 10: Yearly historical performance



Source: SigTech analysis, calculations based on USD returns for the time period January 2000 until January 2022

What's Next for Trend Following?

Trend following strategies have historically proven their value by both delivering attractive risk-adjusted returns and offering effective portfolio diversification. These compelling properties have resulted in steady inflows, which in turn has led to trend-based strategies becoming a more crowded space.

To stay competitive and develop an edge in the trend following market, investors need to build robust, and at the same time flexible, strategies that can be easily adapted to changing market conditions. Those that stay at the forefront of innovation - both in terms of research and operating model - will be the winners.

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